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ORIGINAL ARTICLE

# Evaluating a new wearable lactate threshold sensor in recreational to highly trained cyclists

**Matthew Driller[1] · Nattai Borges[2] · Daniel Plews[3]**

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**Abstract** The determination of a cyclist's lactate threshold (LT) has become an important test performed in sports science laboratories around the world. A limitation of such testing is that it is relatively expensive and invasive, requiring multiple blood samples. The purpose of the current study was to evaluate a commercially available, wearable lactate threshold sensor (WLT) that uses near infrared LED technology to measure gastrocnemius muscle oxygenation and predict LT. The WLT was compared to four traditional calculations of determining LT following an incremental exercise test. Ten male and five female recreational to elite cyclists (mean ± SD; age 24 ± 8, body mass 69.7 ± 7.3 kg, $VO_{2max}$; 59.7 ± 9.9 ml $kg^{-1}$ $min^{-1}$) performed an incremental cycling test to exhaustion. Blood lactate samples were taken at the end of each 3-min stage during the test to determine lactate threshold using four traditional methods (TradLT, $D_{max}$, $mD_{max}$, OBLA). Traditional methods were then compared against the WLT predicted value. The correlation between the WLT and TradLT, $D_{max}$, $mD_{max}$ and OBLA were all $>r = 0.96$. The highest level of agreement for the WLT was with the Dmax method (95 % limits of agreement: ±17 W, TEE = 8.6 W, 4.4 %). The 95 % level of agreement between the WLT and all other traditional methods was <±40 W (TEE <18 W, 8 %). In summary, the WLT is practical, easy to use and exhibits an acceptable level of agreement with four of the traditionally accepted methods of determining lactate threshold.

**Keywords** Blood lactate · Cycling performance · Physiology · D-max · Anaerobic threshold · Aerobic threshold

## 1 Introduction

Monitoring the changes in physiology and performance in cyclists has been studied extensively over the years, leading to a continual development of new, more portable and less invasive monitoring devices used in both laboratory and field settings by sport scientists and coaches. Such devices include power and heart rate monitors, metabolic analysers, GPS/speed systems, and an array of blood analysis equipment. The measurement of blood lactate is one of the more common modes of blood analysis that is performed to monitor cycling performance and physiology. More specifically, the determination of an individual's lactate threshold has become a critical aspect of performance monitoring in cyclists [1, 2]. Indeed, it has been suggested that a combination of lactate threshold and $VO_{2max}$ may be one of the best physiological predictors for endurance cycling performance [3].

The lactate threshold or lactate inflection point can be defined as the exercise intensity in which lactate starts to accumulate exponentially in the bloodstream [4]. This point may also be referred to as the "anaerobic threshold" or "$LT_2$", as opposed to the "aerobic threshold" or "$LT_1$". However, for simplicity, we will refer to this threshold as LT. This threshold occurs when the lactate is produced faster than it can be metabolized during exercise [4]. The most commonly accepted method for measuring LT is determined

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by a lactate profile test, whereby an athlete exercises at increasing intensities/workloads on an ergometer specific to the sport of the athlete being tested, until volitional exhaustion. The blood lactate samples taken at each stage of the test, usually obtained through an ear/finger-prick, are then plotted on a graph against workload/intensity and from this, various methods can be used to determine LT. While there are a number of methods used to determine LT, some of the most commonly used and validated methods include the Dmax and modified Dmax methods, the intersection of two linear splines on a blood lactate curve and the 4 mmol $l^{-1}$ threshold (OBLA) [2, 4–6]. Unfortunately there is no 'gold standard' for which of these methods is used to determine LT, as there are many theories and personal preferences amongst scientists and coaches.

In both the laboratory and in the field, a number of popular bench top and portable blood lactate analysers have been validated [7–10]. While these analysers have all been shown to be valid and reliable, the limitation of all of these methods of assessing blood lactate is that they are invasive, requiring either an ear/finger-prick or even a venous blood sample. This often requires specialist equipment and ongoing consumables, a sterile environment to avoid possible infection, technical expertise and willing athletes, which is not always practical or possible. For this reason, attempts have been made to develop methods that can 'predict' LT using alternative methods that do not require blood sampling. Alternatives used for predicting LT in the literature include heart rate deflection points [11], ventilatory threshold testing [12, 13], rate of perceived exertion [14] and various performance tests [15, 16].

A new product has entered the market which claims to be the worlds first wearable lactate threshold sensor (WLT). The WLT uses near infrared LEDs to emit light into the gastrocnemius muscle and claims to measure relative changes in muscle oxygenation and deoxygenation both at a subcutaneous and superficial level on the lower leg. With the data captured during an incremental exercise test, the WLT software then uses a patented algorithm to predict the LT based on the muscle oxygenation deflection/desaturation point. Our laboratory has determined the inter- and intra- reliability of the WLT device in runners and found it to have a high level of test re-test reliability (CV: <2 %) [17]. However, to date, there have been no published studies investigating the accuracy of this novel method of using a wearable near infrared LED device for predicting LT in cyclists. Therefore, the aim of the current study was to investigate the level of agreement between the new WLT with four other traditional methods of determining LT following an incremental blood lactate profile test in recreational to highly trained cyclists.

## 2 Methods

### 2.1 Subjects

Fifteen recreational to highly trained (international level) cyclists volunteered to participate in the current study. Subjects were 10 males (mean ± SD; age 27 ± 8, body mass 70.7 ± 6.9 kg, $VO_{2max}$; 65.7 ± 7.6 ml $kg^{-1}$ $min^{-1}$) and 5 female (mean ± SD; age 20 ± 3, body mass 67.7 ± 8.3 kg, $VO_{2max}$; 50.1 ± 2.6 ml $kg^{-1}$ $min^{-1}$) cyclists. Both male and female cyclists from a range of abilities and demographics were recruited to test the WLT across a large span of peak power outputs (220–420 W) and lactate threshold powers (WLT-derived values = 197–356 W). Participants were required to provide informed consent before taking part in the study following approval by the institution's Human Research Ethics committee.

### 2.2 Experimental design

The WLT was studied in the controlled setting of a sports science laboratory (21 ± 1 °C) where participants were required to attend a single testing session. The testing session involved subjects performing the incremental $VO_{2max}$/lactate profile test on their own bicycle attached to a cycle ergometer. Subjects were asked to refrain from strenuous exercise (<12 h) and to arrive at each session in a fully rested, hydrated state.

### 2.3 Wearable lactate threshold sensor (WLT)

The wearable lactate threshold sensor used in the current study (BSXinsight multi-sport edition, Texas, USA) was fitted to the subjects on arrival to the laboratory (see Fig. 1). The WLT device was housed in a compression sleeve (provided by the manufacturer) that was fitted so as to ensure that the device was placed on the largest part of the gastrocnemius muscle on the lower leg. The device was approximately 78 mm in height, 37 mm in width and 14 mm in depth and contained three near infrared LEDs (see Fig. 1). The WLT was linked via ANT+ and Bluetooth technology to a smart phone application provided by the manufacturer and also to a Garmin heart rate strap (Garmin Ltd., Kansas, USA) and the cycle ergometer (Wahoo Kickr, Wahoo Fitness, Atlanta, USA). On completion of the exercise test, the WLT software provided a lactate threshold power value in watts (W) that was used for comparison in the current study.

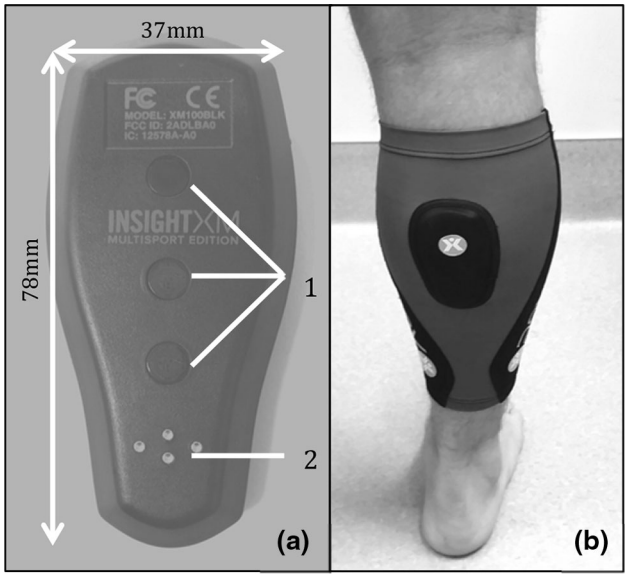


**Fig. 1** **a** The wearable lactate threshold sensor (WLT) and **b** the compression calf sleeve that the WLT it is housed in, positioned over the gastrocnemius calf muscle. *1* Three near infrared LED's, *2* charging connector pins

### 2.4 Incremental exercise test

Participants performed an incremental step test until volitional exhaustion on their own bicycle attached to a magnetically braked cycle ergometer (Wahoo Kickr, Wahoo Fitness, Atlanta, USA), which was calibrated prior to each test according to the manufacturer's instructions. The test protocol was performed according to the WLT manufacturer's instructions, starting at 80–120 W and increased by 20 W every 3 min until the subject could no longer maintain the power output, or cadence dropped below 65RPM. Where subjects failed to complete a stage, peak power output (PPO) was determined using:

$$\mathrm{PPO} = W_{\mathrm{com}} + (t/180 \times 20) \quad (1)$$

where $W_{\mathrm{com}}$ is the power output for the last full workload completed, $t$ is the time in seconds that the final uncompleted workload was sustained, 180 is the target number of seconds in each workload, and 20 is the workload increment in watts, as used previously [18].

Prior to the test, the WLT software required the input of data relating to the individual participants current fitness level. They were required to answer two questions; what is their conversational power output and what is their average power over a 40-km time trial. These questions were required in order to set the starting power output in the test protocol. Cardiorespiratory-metabolic variables were measured throughout the incremental exercise test using a fully automated indirect calorimetry system set at a 15-s sample rate (ParvoMedics TrueOne 2400, Utah, USA). The analyser was calibrated before each test using alpha gases of known concentration, according to the manufacturer's instructions. During the incremental exercise test, each subject was encouraged to give a maximal effort. $VO_{2max}$ was taken as the highest $VO_2$ value recorded over a 1-min period during the incremental test.

### 2.5 Blood lactate sampling

Blood lactate samples were taken in the final 10 s of each stage during the incremental exercise test. At the completion of the test, a further sample was taken 1-min post. Blood lactate concentration was measured via a capillary fingertip sample and analyzed with a Lactate Pro analyzer (Shiga, Japan). The test–retest reliability of the Lactate Pro has been previously reported, with technical error-of-measurement results ranging from 0.1 to 0.4 mmol $L^{-1}$ at blood lactate concentrations of 1–18 mmol $L^{-1}$ [8].

### 2.6 Methods of determining lactate threshold

To compare the WLT values, four different commonly used methods of determining lactate threshold were used; the TradLT, Dmax, mDmax and OBLA methods. While there is no single accepted method of determining LT, all of these methods have been described as being plausible calculations in the research literature [19]. All individual blood lactate values from baseline to 1-min post exercise were added to a previously validated spreadsheet which was used to determine the LT values [19]. The spreadsheet calculated the values using the following definitions.

#### *2.6.1 TradLT*

The point of intersection between two linear splines on a lactate curve. The location of this intersection is estimated by minimizing the sum of the squared differences between the observed lactate values and the fitted values.

#### *2.6.2 $D_{max}$*

Defined as the point that yields the maximal distance from the lactate curve (using a third order polynomial) to the line formed by the lowest and highest lactate values of the curve.

#### *2.6.3 $mD_{max}$*

Defined as the point that yields the maximal distance from the lactate curve (using a third order polynomial) to the line formed between 0.4 mmol $l^{-1}$ above the lowest lactate value and the highest lactate value on the curve.

*2.6.4 OBLA*

Calculated using an inverse prediction by finding the work rate/intensity corresponding to a lactate value equal to a fixed blood lactate concentration of 4 mmol $l^{-1}$.

## 2.7 Statistical analysis

All data are presented as mean ± SD unless otherwise stated. Comparison of the traditionally derived LT values and the WLT values was achieved with Pearson product-moment correlation analysis, standard linear regression, 95 % limits of agreement (LOA), typical error of estimate (TEE) and mean bias (%). While correlation analysis indicates the degree to which two variables are associated, it does not necessarily indicate the extent to which values agree or disagree. To overcome this limitation, the approach of quantifying the level of agreement between the different LT methods measuring the same parameter (in this case lactate threshold) was employed [20, 21]. The mean difference between methods (2 standard deviations or 95 % of a normally distributed population) was determined. Inspection of the slope and intercept of the linear regression was also undertaken to characterize the level of agreement between methods. The TEE and mean bias between the WLT and other methods was determined using an excel spreadsheet [22], with the TEE expressed both in raw units and as a percentage. The magnitude of correlation between the WLT and the other common methods of LT assessment was assessed with the following thresholds: <0.1, trivial; <0.1–0.3, small; <0.3–0.5, moderate; <0.5–0.7, large; <0.7–0.9, very large; and <0.9–1.0, almost perfect. If the 90 % confidence intervals (CI) overlapped small positive and negative values, the magnitude of correlation was deemed 'unclear'.

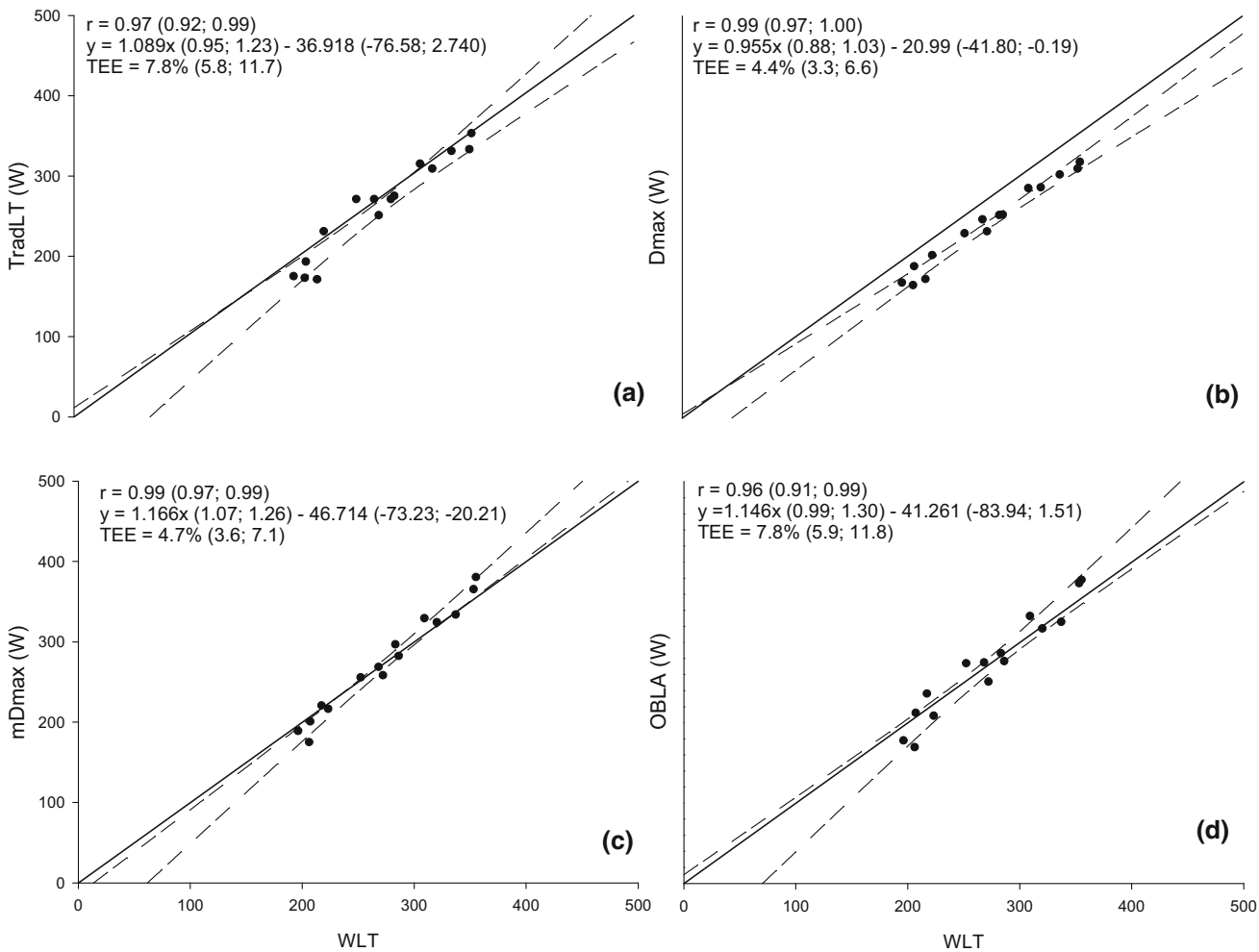


**Fig. 2** The correlation plots (±95 % confidence intervals expressed by *dashed lines*) and linear regression equations for: **a** WLT vs TradLT; **b** WLT vs $D_{max}$; **c** WLT vs $mD_{max}$; **d** WLT vs OBLA. *Solid black line* represents line of equivalence ($r = 1.0$)

## 3 Results

The results for each lactate threshold method comparison are presented in two graphical formats. Firstly, each comparison is illustrated by correlational analysis, which shows the linear regression equation, the correlation coefficient and the 95 % confidence intervals (Fig. 2: WLT vs TradLT; WLT vs Dmax; WLT vs mDmax; WLT vs OBLA). Secondly, the level of agreement (±95 % limits of agreement) between the WLT and each of the other methods is shown in Fig. 3 (WLT vs TradLT; WLT vs Dmax; WLT vs mDmax; WLT vs OBLA).

The correlations between the WLT and the TradLT, $D_{max}$, $mD_{max}$ and OBLA methods were $r$ (90 % confidence interval) = 0.97 (0.92; 0.99), 0.99 (0.97; 1), 0.99 (0.97; 1) and 0.96 (0.9; 0.98), respectively. The 95 % limits of agreement between WLT and each lactate threshold method was less than 40 W over the range of lactate threshold powers (197–356 W, Table 1). The highest level of agreement and lowest TE for the WLT was observed with the Dmax method (LOA; ±17 W, TE; 8.6 W, 3.2 %). The lowest level of agreement and TE was between WLT and the OBLA method (LOA; ±37.0 W, TE; 17.7 W, 6.5 %).

## 4 Discussion

The results obtained in this evaluation support the use of the wearable lactate threshold sensor as a method used by athletes and coaches to monitor changes in lactate threshold. The very large correlations ($r > 0.96$) between the WLT and other methods of determining lactate threshold through the range of 195–356 W, coupled with the

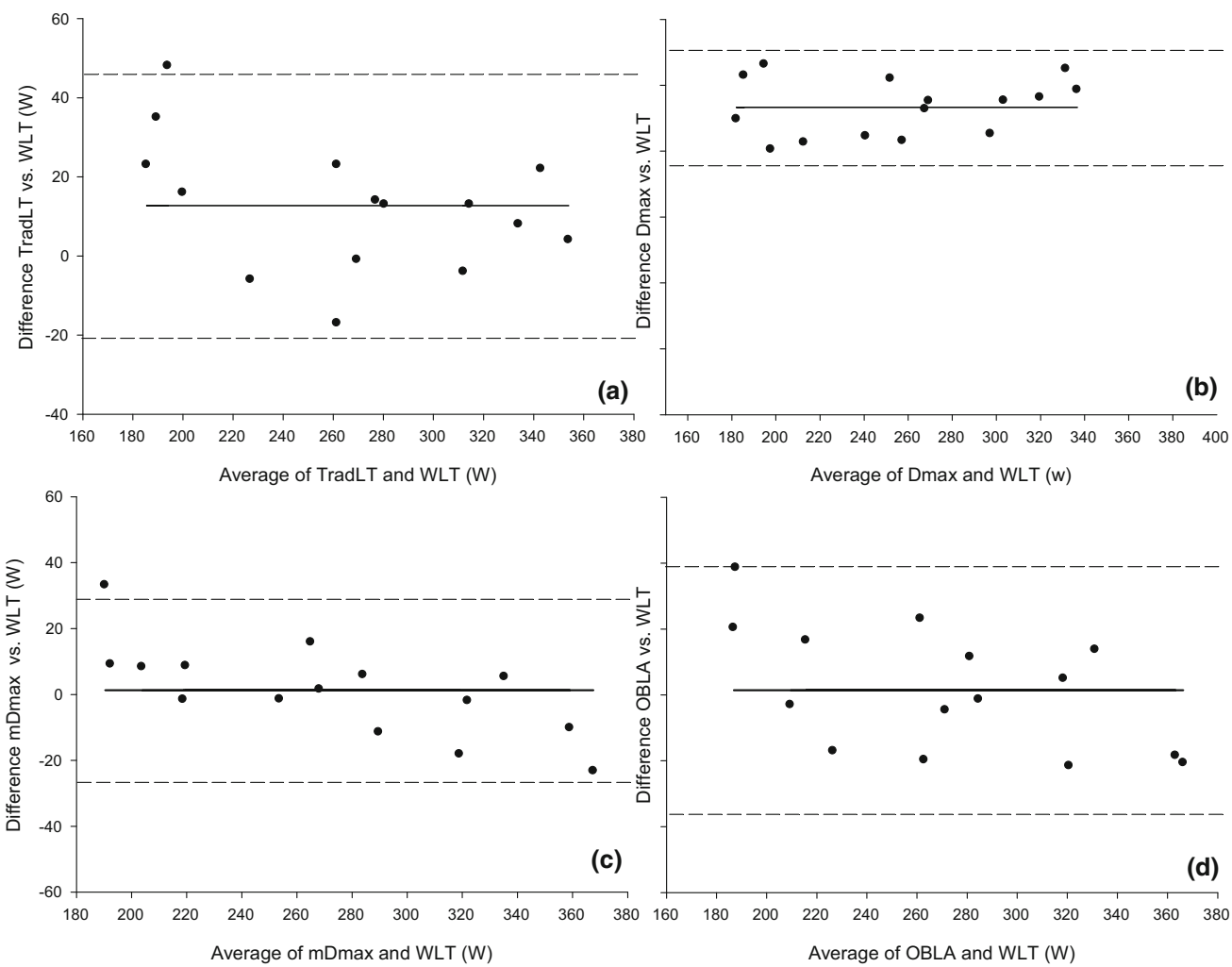


**Fig. 3** The level of agreement plots (Bland–Altman) showing ± 95 % limits of agreement (represented as *dashed lines*) between the WLT and each of the other methods: **a** WLT vs TradLT; **b** WLT vs $D_{max}$; **c** WLT vs mDmax; **d** WLT vs OBLA. *Solid black line* represents the mean difference between methods

**Table 1** Comparison of the WLT with four other methods of determining lactate threshold, showing the mean difference, the range of mean differences (limits of agreement), mean bias, the typical error of estimate (W and %) and the Pearson's product-moment correlation (*r*) between each pair

| WLT compared with | Mean difference ± SD (W) | Range of mean difference (±2SD) (W) | Mean bias (%) | TEE* (W) | TEE* (%) | Correlation (*r*) |
|---|---|---|---|---|---|---|
| TradLT | 12.7 ± 16.6 | −20.5 to 45.9 | 6 | 16.5 | 7.8 | 0.97 |
| $D_{max}$ | 33.3 ± 8.7 | 15.9 to 50.7 | 15.5 | 8.6 | 4.4 | 0.99 |
| $mD_{max}$ | 1.3 ± 14.0 | −26.6 to 29.2 | 1.3 | 11.0 | 4.7 | 0.99 |
| OBLA | 1.5 ± 18.9 | −36.2 to 39.2 | 1.5 | 17.7 | 7.8 | 0.96 |

*2SD* two standard deviations

* Typical error of estimate

relatively low typical error (~4 %, ~10 W) between WLT and other traditional methods, underline the suitability of the WLT in a practical setting.

The purpose of the WLT is to predict the lactate threshold (more commonly known as "anaerobic threshold" or "$LT_2$") that corresponds to intensities where lactate production begins to exceed the maximal rate of blood lactate removal [4]. This threshold is closely related to maximal power output cyclists can maintain for ~60 min which is dependent on the training status of the individual [23]. Indeed, the power associated with LT is highly correlated with performance in cycling [24, 25], more so than other physiological determinants such as $VO_{2max}$ [25]. Accordingly, the ability to accurately determine LT is an important consideration as it allows for coaches and athletes to; (1) monitor progression in training/performance, and (2) define accurate intensities for training prescription. Moreover, WLT devices such as the one assessed in the current study, may allow for more routine assessments of athletes without the associated costs of lactate profile tests in sport science laboratories.

The company that manufactures the WLT does not report which of the many different methods of determining lactate threshold was used for the development of their patented algorithm. Given the range of LT values that can be produced by the different traditional methods, coupled with the fact that there is no known "gold standard" for LT, it makes it difficult to conclude whether the WLT provides a valid prediction of LT. Indeed, the four traditional methods used in the current study produce a range of mean LT values between 240 and 273 W. Therefore, the reliability of the WLT is perhaps the most important factor when it comes to using the device to monitor athletic performance. While the reliability was not investigated in the current study, observations from our laboratory in runners have shown the WLT to be highly reliable, with a test–retest coefficient of variation of <2 % [17].

Previous studies investigating the suitability of muscle oxygenation breakpoints using near-infrared spectroscopy (NIRS) systems to determine exercise thresholds have shown similar results to those of the present study [26–28]. Belloti et al. (2013) reported high correlations ($r = >0.76$) for $VO_2$ and heart rate values derived from change in deoxyhemoglobin concentration from the vastus lateralis muscle during a maximal lactate steady state test on a cycle ergometer [28]. Wang et al. (2006) compared threshold workload (W), $VO_2$ and heart rate values from multiple NIRS parameters (change in oxyhaemoglobin concentration, change in deoxyhemoglobin concentration) also from the vastus lateralis muscle with those derived from LT during incremental cycling exercise. The authors reported that the NIRS parameter with the strongest correlations to LT was the change in deoxyhemoglobin concentration [29].

It is unknown what an acceptable level of agreement or typical error of measurement is for predicting lactate threshold. However, a typical error of measurement of <15 % has been deemed to be acceptable for blood lactate measurement using a portable analyser [9]. Therefore, given all of the traditional LT values had a TEE lower than 15 % when compared to the WLT, we can assume that this is an acceptable level of agreement.

To conclude, the findings of our study suggest that the WLT is a practical tool for estimating lactate threshold workloads for cyclists. We found acceptable levels of agreement between the WLT and four of the commonly used blood lactate threshold methods. The WLT could be implemented by coaches and athletes as a portable and non-invasive method to monitor lactate threshold workload changes in cyclists of all abilities.

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